GroupTrack

User Manual

Version 2.4

Off-Road Convoy Tracking for Android

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grouptrack.org

Facebook: GroupTrack Off Road Navigation

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Chapter 1

What is GroupTrack?

GroupTrack is an Android application for off-road GPS tracking and convoy coordination. It works in two modes: as a standalone GPS tracker for solo riders (no radio required), and as a convoy tracking system using Meshtastic mesh radios to share real-time GPS positions between riders in a group.

In convoy mode, everyone sees where everyone else is on an offline map with no cell service required. Whether you ride solo and want free offline maps with GPS track recording, or you coordinate group rides through canyon country and need real-time position sharing, GroupTrack is built for you.

1.1 Key Features

- Real-time convoy tracking via Meshtastic mesh radio network
- Offline satellite and topographic maps (Esri tile sources)
- GPS track recording with automatic GPX file saving
- Trail overlay data for OHV route visualization
- Background map tile downloads with notification progress
- Download queue management with concurrent processing
- Planning Map for download management and ride planning
- Track file management with import, rename, delete, share, and date correction
- Lead and tail cart identification with convoy spread monitoring
- AutoPause and Auto-Wake for fuel stops and camp
- Solo mode for independent GPS tracking without radio
- Compatible with T1000-E and other Meshtastic radios

1.2 What You Need

Required Hardware

- Android device: Android 10 or later. Samsung and Pixel devices tested. Name-brand phones and tablets recommended for reliable Bluetooth.
- Map tiles: Download offline map tiles over WiFi before your ride. Without tiles, the map shows a blank background in areas without cell service.

Required for Group/Convoy Mode

- Meshtastic radio: T1000-E or compatible Meshtastic device. One per rider. Must be on the same frequency, channel, and modem preset as other riders in the convoy.
- Bluetooth: The app connects to the radio via Bluetooth Low Energy (BLE). The radio must stay within BLE range (approximately 50 feet) of your phone at all times during the ride.

CRITICAL: Your T1000-E radio MUST remain on firmware version 2.6.11. Do NOT accept firmware update prompts. Newer firmware versions destabilize BLE connectivity and radio configuration.

CRITICAL: Keep your T1000-E in your cart at all times during rides. Do NOT remove it or carry it away from your phone. Exceeding 50-foot BLE range causes Bluetooth supervision timeout disconnections that disrupt position tracking.

Budget Android tablets (Tab8NEU and similar) have known Bluetooth issues with T1000-E radios. The BLE connection drops after 8-10 seconds. This is a confirmed hardware limitation. Use a name-brand phone or tablet for reliable operation. **See Appendix A for tested devices.**

1.3 Solo vs Group Mode

GroupTrack automatically detects whether a Meshtastic radio is connected. With a radio connected, the app operates in Group Mode with full convoy tracking. Without a radio, it operates in Solo Mode with GPS tracking, offline maps, and track recording but no convoy features.

Solo Mode Features

- GPS position tracking on offline maps
- Track recording with GPX file saving
- Trail overlay visualization
- All offline map tile features (download, manage, display)
- Track file management (import, rename, delete, share)

Group Mode Features (everything in Solo Mode plus)

- Real-time position sharing via mesh radio
- Lead and tail cart identification
- Convoy spread monitoring
- AutoPause/Auto-Wake for group stops
- Radio configuration tools

Chapter 2

Getting Started

2.1 Install GroupTrack

GroupTrack is available through the Google Play Store internal testing program. If you received a Google Play internal test invite, follow the installation instructions in the Tester Installation Guide provided by your ride organizer.

If you received a direct APK file for side-loading, see the side-loading instructions provided by your ride organizer. Note that Play Store and side-loaded versions use different signing keys and cannot be installed over each other. Uninstall one before installing the other.

If upgrading from a previous version, all your downloaded map tiles, saved tracks, and settings are preserved automatically. You do not need to re-download maps.

2.2 Grant Permissions

GroupTrack requires several permissions to function correctly. The app will prompt you for each permission on first launch or when a feature requires it.

Location Permission

1. When prompted, tap "Allow While Using the App" first.
2. The app will then explain why it needs background location access and ask you to grant "Allow All the Time."
3. This opens your device Settings. Find GroupTrack in the list and select "Allow all the time."
4. Return to the app.

If you only grant "While using the app," GPS track recording will stop when you turn off your screen or switch apps. Your ride track will be empty. You MUST grant "Allow all the time" for track recording to work.

Storage Permission

1. The app will request "All Files Access" permission.
2. This opens your device Settings. Find GroupTrack and toggle on "Allow access to manage all files."
3. Return to the app.

This permission is required for offline map tile storage, track file management, and GPS recording.

Notification Permission

1. Grant notification permission when prompted.
2. Required for background download progress notifications and download completion alerts.

Without notification permission, map tile downloads will still work but you will not see progress updates in your system tray.

Bluetooth Permission

Required for connecting to your Meshtastic radio. Grant when prompted. Without this permission, GroupTrack operates in Solo Mode only.

Nearby Devices Permission

Required for Bluetooth device discovery (scanning for available radios). Grant when prompted.

If you accidentally deny a permission or revoke it later in Settings, GroupTrack will prompt you again on the next app resume. You do not need to reinstall.

2.3 Connect to Your Radio (Group Mode)

1. Power on your Meshtastic radio (T1000-E or compatible).
2. In GroupTrack, you will see a Bluetooth connection prompt or a radio list.
3. Select your radio from the list. The app connects via BLE.
4. Wait for the connection indicator to turn green (may take 10-15 seconds).
5. Once connected, your position and other riders' positions will appear on the map.

GroupTrack remembers your last connected radio and will automatically reconnect on future launches. Keep your radio powered on and within 50 feet of your phone throughout the ride.

2.4 First Launch Checklist

- Install GroupTrack from Play Store or APK
- Grant all permissions (Location, Storage, Notifications, Bluetooth, Nearby Devices)
- Power on T1000-E radio (verify firmware 2.6.11)
- Connect radio in GroupTrack (green indicator)
- Open Planning Map and download offline tiles for your ride area
- Verify downloaded area coverage (toggle Downloaded Areas in Display panel)
- You are ready to ride

Chapter 3

The Convoy Map

The convoy map is GroupTrack's primary screen for live ride operations. It shows your position, all connected riders in the convoy, and the mesh radio status. The convoy map is optimized for real-time situational awareness during rides.

3.1 Map Display

- Your cart ("My Cart") is shown on the map with your position updated in real time via GPS.
- Other riders in the convoy appear as separate markers with their callsigns, updated as the mesh network relays GPS data.
- The map uses satellite tiles by default with road and place name label overlays.
- Trail overlays display known OHV trails when toggled on in the Display Panel (currently southern Utah coverage).
- AutoPan is off by default. During recording, AutoPan turns on to follow your position. When you stop recording, it turns off so you can pan and zoom freely.
- Lost rider alerts appear when a rider drops off the mesh network.

3.2 Map Toolbar

The convoy map has a fixed toolbar at the top of the screen. The toolbar is positioned to the right of the RECORD button and provides the following controls:

- PLAN button: one-tap navigation to the Planning Map for downloading tiles and planning rides.
- SAT / TOPO / TOPO+ buttons: switch between satellite imagery, topographic, and USGS topo maps. The active source is highlighted in blue.
- NET / LOCAL toggle: switch between online tiles (requires internet) and locally cached tiles (works offline).

See Chapter 5 for complete toolbar documentation.

3.3 Display Panel

A floating DISPLAY panel provides toggle controls for map layers. Tap the DISPLAY pill to expand it. The panel is draggable -- hold and drag to reposition it anywhere on the map. See Chapter 6 for complete display panel documentation.

3.4 Lead Cart Selection

When you start a group recording, a dialog appears asking you to select the lead cart. Tap the callsign of the rider at the front of the convoy. GroupTrack uses the lead cart to compute convoy positions and heading. The tail cart is identified automatically as the rearmost rider.

In solo mode, lead cart is assigned automatically (you are the only rider).

3.5 Cart Position and Tracking

GroupTrack determines each cart's position with a four-tier fallback chain:

- Live radio GPS (preferred): the cart's radio reports its current GPS fix.

- Last known radio position: if the radio is currently silent but reported recently.
- Phone GPS: for the user's own cart, if the radio cannot be reached, the phone's GPS is used.
- Trailer cart substitution: for on-track carts that have gone silent, the cart is shown at the position of the furthest trailing cart still reporting. This keeps silent carts visible on the map.

Carts that have deviated off the planned track keep their own last known position so operators can locate them.

3.6 Speed Display

Your current speed (mph) is displayed on the map HUD after a brief warmup period. During the first 60 seconds after starting a ride, the speed display may show 0 mph while the GPS acquires enough position samples to compute accurate speed.

Mileage traveled (trip odometer) is not displayed in V2.4. This feature is planned for V2.5.

3.7 The GroupTrack Menu

Tap the menu button to open the GroupTrack menu. It has two accordion sections:

EVENT / RIDE

- Create rides and manage ride configurations
- Transfer ride configs between devices
- Import ride files
- Apply radio configuration for convoy use
- Restore prior radio configurations

MAP SETTINGS

- Work With Tracks: full track file management and import
- Planning Map: independent map for downloads, tracks, and ride planning
- Cloud map sync (coming in V2.5)

Long-press the GroupTrack header for developer settings (password-protected).

3.8 Navigating to Planning Map

Tap the PLAN button in the toolbar to switch to the Planning Map. The Planning Map is your tool for downloading offline tiles, viewing trail data, managing tracks, and planning rides. Tap CONVOY on the Planning Map toolbar to return to the convoy map.

Chapter 4

The Planning Map

The Planning Map (formerly Map Viewer) is your central tool for downloading offline map tiles, viewing trail data, managing tracks, and planning rides. It provides an online map with full download management capabilities.

Use the Planning Map before your ride to prepare offline maps for your ride area. After the ride, use it to review recorded tracks and manage your track library.

4.1 Accessing the Planning Map

- From the convoy map: tap PLAN in the toolbar (primary access)
- From the GroupTrack menu: expand MAP SETTINGS, tap Planning Map
- To return: tap CONVOY in the Planning Map toolbar

4.2 Search

The Planning Map has a search field always visible at the top of the screen. Type a city, park, trail name, or geographic feature and tap FIND to fly to that location.

- Results appear as a dropdown below the search field
- Tap any result to navigate the map to that location
- The map zooms to an appropriate level for the area
- Search works with city names, park names, trail names, coordinates, and general geographic terms

Search requires an internet connection. Download your tiles after searching, then switch to LOCAL mode for offline use.

4.3 TRACKS Button

The Planning Map header includes a TRACKS button for quick access to track management. Tap TRACKS to open the track panel where you can:

- Browse and search all saved track files
- Select individual tracks to display on the map
- Fit the map to show all loaded tracks
- Clear all loaded tracks from the map
- Access track actions: rename, delete, share, move to downloads, fix creation date
- Import tracks from your Downloads folder

The TRACKS button refreshes the file list each time you open it, so newly imported or recorded tracks appear immediately.

4.4 Downloading Tiles

To download offline map tiles for your ride area:

1. Navigate to your ride area using the search field.
2. Tap the DOWNLOAD button in the toolbar.
3. Adjust the download zoom level slider (higher zoom = more detail, larger download). z14 for wide area, z15-18 for trail detail.
4. Tap "DOWNLOAD REGION" to activate draw mode.
5. Draw a rectangle around the area you want to download.
6. Confirm the download. It enters the queue and starts automatically.
7. A notification appears in your system tray showing progress.
8. You can close GroupTrack. The download continues in the background.

Download maps on WiFi before your ride. A typical ride area (20-30 miles of trail) takes 5-15 minutes depending on zoom level and connection speed. Download z14 first for wide coverage, then z15-18 for just the trail system.

4.5 Download Panel Controls

The DOWNLOAD button expands to reveal additional controls:

- Download Region: draw a rectangle to download tiles for that area
- Remove Area: draw a rectangle to delete tiles for that area (coming in V2.5)
- Download Zoom slider: controls the maximum zoom level for downloads (16-19)
- Fly Over Radius slider: controls the zoom level when flying to search results

4.6 Verifying Downloads

After downloading tiles, verify your coverage:

1. Tap the DISPLAY pill to open the display panel.
2. Toggle "Downloaded Areas" ON.
3. Blue shading appears on the map where you have offline tiles.
4. Verify your ride area is fully covered in blue.
5. Switch to LOCAL mode using the NET/LOCAL toggle to confirm tiles render correctly offline.
6. Switch back to NET mode when done verifying.

4.7 Map Toolbar on Planning Map

The Planning Map shares the same toolbar design as the convoy map (see Chapter 5). The CONVOY button replaces the PLAN button for returning to the convoy map. The SAT/TOPO/TOPO+ and source highlight work identically.

4.8 Returning to Convoy Map

Tap CONVOY in the Planning Map toolbar to return to the convoy map. Your map position and settings are preserved when switching between maps.

Chapter 5

Map Toolbar

Both the convoy map and Planning Map share a consistent, compact toolbar at the top of the screen. The toolbar provides tile source selection, online/offline mode switching, and quick navigation between maps.

5.1 Toolbar Layout

The toolbar is a single fixed bar with the following elements from left to right:

Navigation Button

- On the convoy map: "PLAN" -- tap to open the Planning Map
- On the Planning Map: "CONVOY" -- tap to return to the convoy map

One-tap navigation between maps. No menu diving required.

Tile Source Buttons

Three buttons for selecting the active tile source:

- SAT: Esri World Imagery -- high-resolution satellite and aerial photography with road and place name labels.
- TOPO: Esri World Topo Map -- topographic map with terrain shading, contour indicators, roads, trails, and place names.
- TOPO+: Esri USA Topo Maps -- USGS-style topographic maps with detailed contour lines, elevation markers, and traditional topo symbology.

The active source is highlighted in blue. Tap any button to switch sources instantly. Both online display and offline tiles switch together.

NET / LOCAL Toggle

A switch at the right end of the toolbar controls the tile source:

- NET (green label): tiles loaded from Esri servers in real-time. Best quality and most current imagery. Requires an active internet connection.
- LOCAL (blue label): tiles loaded from your downloaded cache on device. Works completely offline. Must be downloaded in advance using the Planning Map.

Switch to LOCAL before entering areas without cell service. Switch to NET when you have connectivity for the best tile quality and coverage.

5.2 Toolbar Positioning

On the convoy map, the toolbar is positioned to the right of the RECORD button to avoid overlap. On the Planning Map, the toolbar sits directly below the search field with minimal gap.

Chapter 6

Display Panel

The Display Panel is a floating, draggable toggle panel that controls map layer visibility. It appears on both the convoy map and Planning Map with the same options, providing consistent layer control across both views.

6.1 Opening and Closing

- Tap the "DISPLAY" pill to expand the panel and show all toggles.
- Tap "DISPLAY" again to collapse the panel back to a compact pill.
- The panel remembers its expanded/collapsed state within the current session.

6.2 Dragging and Positioning

The Display Panel is draggable. Touch and hold the panel header, then drag it to any position on the map. This lets you move it away from areas you need to see. Useful when the panel covers a trail junction or area of interest.

The panel position resets when you navigate between screens (convoy to Planning Map and back). Position persistence is planned for V2.5.

6.3 Layer Toggles

The Display Panel includes the following layer toggles:

Tracks

Toggle ON to load and display all saved GPS tracks from your track library on the map. Tracks appear as neon green polylines. The first time you toggle tracks ON, GroupTrack reads all track files from Documents/my_tracks/ and loads them sequentially. This may take a few seconds for large track libraries (68+ files). Subsequent toggles show/hide the already-loaded tracks instantly.

Toggle OFF to hide all tracks from the map without removing the loaded data.

Trails

Toggle ON to load and display trail overlay data on the map. Currently includes Utah St. George area trails covering OHV routes, multi-use trails, and hiking paths. Trails appear as colored overlay lines showing established paths.

The first time you toggle trails ON, the trail data is loaded from the app assets. This may take a moment for large datasets. Once loaded, toggling is instant.

Toggle OFF to hide trail overlays and declutter the map.

Waypoints (V2.5)

Greyed out in V2.4. Coming in V2.5 -- will toggle visibility of user-created waypoints including trailheads, fuel stops, gates, hazards, scenic viewpoints, water sources, camps, and parking areas.

Routes (V2.5)

Greyed out in V2.4. Coming in V2.5 -- will toggle visibility of planned routes including drawn routes, snap-to-trail routes, and imported GPX/KML route files.

Downloaded Areas

Toggle ON to show blue shading on the map where you have downloaded offline tiles. This helps you verify which areas are covered for offline use and which areas still need downloading.

The scan reads your tile cache directory and calculates coverage boundaries. For large downloaded areas, the scan may take a few seconds. The blue overlay updates each time you toggle ON.

Toggle OFF to remove the blue shading overlay.

Downloads in Progress (V2.5)

Coming in V2.5. Will show colored shading for active download queue entries: pink for areas being downloaded, red for areas being removed.

Chapter 7

Recording a Ride

7.1 Start Recording

1. Tap the RECORD button on the convoy map screen.
2. If prompted for location permission, grant "Allow all the time" as described in Section 2.2.
3. Solo mode: Recording starts immediately. Lead cart is you.
4. Group mode: A dialog appears. Select the lead cart by tapping their callsign.
5. The map immediately centers on your cart position. No delay.

Your cart identity is locked at RECORD time. It will not change or flicker during the ride, even if other carts join the convoy after recording starts.

7.2 During the Ride

- Your GPS position is recorded every 3 seconds and saved to a track file.
- The track line draws on the map as you move, showing your ride path in real time.
- Other riders' positions update via the mesh radio network.
- AutoPan follows your cart while recording. You can temporarily drag the map; it re-centers on the next GPS update.
- You can lock and unlock your phone -- recording continues in the background (requires "Allow all the time" location permission).
- The recording indicator remains visible on screen throughout the ride.

7.3 AutoPause and Auto-Wake

When your convoy is stationary for approximately 10 minutes within roughly 50 feet of one spot, GroupTrack automatically pauses to save battery:

- GPS updates pause to conserve battery
- Radio Bluetooth disconnects
- The screen displays a flashing "ZZZ" indicator

To wake: tap the ZZZ indicator. GroupTrack reconnects to your radio, resumes GPS updates, and rejoins the convoy. This feature is ideal for fuel stops, lunches, and overnight camps.

7.4 Stop Recording

1. Tap the STOP button.
2. A dialog appears asking you to name the track.
3. Enter a descriptive name (e.g., "Bar 10 Trail April 2026").
4. Confirm. The track is saved as a GPX file in Documents/my_tracks/ on your device.

After stopping, AutoPan turns off so you can pan and zoom the map freely to review your ride.

Chapter 8

Work With Tracks

GroupTrack keeps all your imported and recorded tracks in Documents/my_tracks/ on your device. The Work With Tracks screen provides full track file management including browsing, searching, sorting, renaming, deleting, sharing, and importing.

8.1 Accessing Work With Tracks

- From the GroupTrack menu: expand MAP SETTINGS, tap Work With Tracks
- From the Planning Map: tap the TRACKS button, then choose Work With Tracks from the dropdown

8.2 Browsing and Searching

- Search: type into the search box at the top. The list filters live as you type.
- Sort: tap the date/name button to cycle through four modes: Date Newest, Date Oldest, Name A-Z, Name Z-A. Default is Date Newest.
- Filter tabs: All (every file), Saved (completed tracks), In-Progress (orphaned temp files from interrupted recordings).

8.3 Track Actions

Tap any track to open its action menu:

- Rename: change the file name. The .gpx or .kml extension is preserved automatically.
- Delete: permanent removal. Confirms before acting.
- Share: opens Android's share sheet. Attach to email, send to Drive, AirDrop, etc.
- Move to Downloads: copies the file to your Downloads folder for backup or transfer. The original stays in my_tracks.
- Fix Creation Date: updates the file's date to match the earliest GPS timestamp inside the file. The date column updates immediately without leaving the screen.

8.4 Bulk Fix Dates

The Fix Dates button next to the track count rewrites date stamps for every visible track at once. Use this once to clean up tracks imported before V2.4 -- they will start showing their real recording dates instead of import dates.

> *GroupTrack never deletes your tracks automatically. Cleanup of orphan in-progress files is a deliberate user action, never automatic.*

8.5 Import Tracks from Downloads

Tap the Import button in the Work With Tracks header to open the import screen. GroupTrack scans your Downloads folder for .gpx and .kml files.

- Each file is shown with filename, file size, date, and type (GPX/KML).
- Select files individually with checkboxes, or tap Select All.

- Tap Import N Selected. A progress dialog shows each file being processed.
- Multi-track files (like onX bundles with many tracks in one GPX) are automatically split into individual track files.
- Each imported track gets its date corrected to the original GPS recording date.
- A recap dialog shows imported count, skipped (already exist), and failed count with track names.

Also accessible from Planning Map: tap the TRACKS button, then choose Import from Downloads.

Chapter 9

Track Display

GroupTrack provides two ways to display saved tracks on the map: the Display Panel toggle for loading all tracks at once, and the track picker for selecting individual tracks.

9.1 Display Panel Track Toggle

Toggle Tracks ON in the Display Panel to load ALL saved tracks from your track library onto the map. Tracks appear as neon green polylines.

- First toggle ON: reads all track files from Documents/my_tracks/ sequentially. Each file is read on a background thread, parsed (GPX or KML), and loaded onto the map via JavaScript. This may take a few seconds for large libraries.
- Subsequent toggle ON: shows tracks instantly without re-reading files from disk.
- Toggle OFF: hides all tracks from the map.

Each map (convoy and Planning) loads tracks independently. Loading tracks on one map does not affect the other map.

9.2 Track Picker (Planning Map)

For more granular control, use the track picker on the Planning Map:

1. Tap the TRACKS button in the Planning Map header.
2. The track panel opens showing all saved track files.
3. Tap any track to load it on the map. A colored indicator shows loaded tracks.
4. Tap a loaded track again to unload it.
5. Use the search box to filter by name.
6. Tap FIT to zoom the map to fit all loaded tracks.
7. Tap CLEAR to remove all loaded tracks from the map.

Each track in the picker has an overflow menu for actions: rename, delete, share, move to downloads, and fix creation date.

9.3 Track Colors

Tracks loaded via the Display Panel toggle appear in neon green (#39FF14), the GroupTrack signature color. Tracks loaded individually from the picker use a rotating color palette for visual distinction between overlapping routes.

Chapter 10

Download Queue

GroupTrack V2.4 introduces a background download queue for map tile downloads. You can queue multiple download areas, close the app, and downloads continue running via Android's WorkManager background service.

10.1 How the Queue Works

- When you confirm a download in the Planning Map, it enters the download queue.
- The queue processes up to 2 downloads simultaneously.
- Each download fetches tiles for all active tile sources (SAT, TOPO, TOPO+) and their overlay layers (labels, transport, places).
- Downloads survive app backgrounding, screen rotation, device sleep, and even app kills.
- If the device restarts, pending downloads resume automatically.
- Download progress is tracked per-tile with periodic state saves.

10.2 Notifications

When a download is active, a persistent notification appears in your Android system tray:

- Shows the download label, current tile count, and progress percentage.
- Includes a CANCEL button to stop the download directly from the notification.
- Stays visible even when GroupTrack is closed or in the background.
- A completion notification appears when a download finishes, showing the total tile count.
- Failed downloads show an error notification with the failure reason.

10.3 Queue Panel

The queue panel appears at the bottom of the Planning Map when downloads are active. Tap the summary pill to expand it and see full details.

Summary View (Collapsed)

Shows the number of active and queued downloads with a spinning progress indicator. One tap expands to the detail view.

Detail View (Expanded)

- Active downloads: individual progress bar per download with label, tile count, percentage, and Cancel button.
- Queued items: waiting to start, showing position in queue and estimated tile count.
- Completed items: finished downloads with final tile counts.
- CANCEL ALL button: stops all active and queued downloads at once.
- CLEAR DONE button: removes completed entries from the list.

10.4 Download Priority

Downloads are processed in priority order:

- Priority 1: Ride enrollment auto-downloads (V3.0 feature)
- Priority 3: User-initiated downloads (standard priority for manual tile downloads)
- Priority 7: Background tile source refresh after changing sources

Higher-priority downloads are processed first. Active downloads are not interrupted -- the next available slot goes to the highest-priority queued item.

Chapter 11

Offline Map Tiles

GroupTrack works best with pre-downloaded map tiles. Without them, the map shows a blank background in areas with no cell service. The tile download system stores tiles locally on your device for offline access.

11.1 Why Download Tiles?

Off-road riding typically takes you to areas with no cell service. Without pre-downloaded tiles, the map cannot display any imagery or terrain data. Downloading tiles before your ride ensures you have full map coverage regardless of cell service availability.

11.2 Downloading Tiles Step by Step

1. Connect to WiFi before your ride (recommended for faster downloads).
2. Open the Planning Map (tap PLAN on the convoy map toolbar).
3. Use the search field to navigate to your ride area.
4. Tap the DOWNLOAD button to open the download panel.
5. Set the download zoom level: z14 for wide area coverage, z15-18 for trail-level detail.
6. Tap DOWNLOAD REGION and draw a rectangle around your ride area.
7. Confirm the download. It enters the queue and starts automatically.
8. Wait for completion or close GroupTrack -- downloads continue in the background.

Download z14 for the whole region first, then z15-18 for just the trail system. A typical ride area (20-30 miles of trail) requires 200-500 MB of tile storage. Both convoy map and Planning Map share the same tile cache.

11.3 Tile Storage

Downloaded tiles are stored on your device at /sdcard/Documents/GroupTrack/maps/tiles/ in separate directories per source and layer. The directory structure is: source/zoom/x/y.png.

- Tiles persist across app updates.
- Tiles are not affected by clearing app data.
- Both the convoy map and Planning Map read from the same tile cache.
- The only ways to remove tiles are: the Remove Area function (V2.5), or manually deleting the tile directories via a file manager.

11.4 Verifying Downloaded Areas

Use the Downloaded Areas toggle in the Display Panel to verify your tile coverage. Blue shading shows where you have offline tiles stored on disk. Gaps in the blue coverage indicate areas that need downloading.

Chapter 12

Tile Sources

GroupTrack V2.4 includes three default tile sources from Esri, the leading provider of geographic information systems and mapping data.

12.1 SAT (Esri World Imagery)

High-resolution satellite and aerial photography covering the entire globe. This is the default tile source and provides the most familiar view for off-road navigation.

- Best for: identifying terrain features, vegetation types, water bodies, structures, fence lines, and roads.
- Includes: road name labels and place name overlay layers for reference.
- Zoom levels: clear imagery from wide area (z10) down to trail-level detail (z18).

12.2 TOPO (Esri World Topo Map)

Topographic map with terrain shading, contour indicators, roads, trails, and place names. Provides a traditional map view with elevation context.

- Best for: understanding elevation and terrain profile while maintaining readable labels and trail markings.
- Includes: contour shading, trail markings, road classifications, land use boundaries.
- Useful when: satellite imagery is too busy or you need clear trail/road distinction.

12.3 TOPO+ (Esri USA Topo Maps)

USGS-style topographic maps with detailed contour lines, elevation markers, and traditional topo map symbology. The most detailed terrain data available.

- Best for: precise elevation planning, route difficulty assessment, identifying drainage patterns and ridgelines.
- Includes: full USGS contour lines, benchmark elevations, quad boundaries.
- Useful when: you need exact terrain information for technical route planning.

12.4 Online vs Offline Modes

- NET (Online): tiles fetched from Esri servers in real-time. Provides the best quality and most current imagery. Requires an active internet connection (WiFi or cellular data).
- LOCAL (Offline): tiles loaded from your downloaded cache. Works without any internet connection. Must be downloaded in advance using the Planning Map. Quality matches what was downloaded.

The NET/LOCAL toggle on the toolbar switches between modes instantly. Always download tiles and verify coverage in LOCAL mode before heading to areas without cell service.

Chapter 13

Radio Configuration

GroupTrack includes tools to configure your Meshtastic radio for convoy use. Proper radio configuration ensures all riders can communicate via the mesh network.

13.1 Recommended Settings

The recommended convoy configuration is LONG_FAST mode, which provides 3-5 mile range with good data throughput for GPS position updates. This mode balances range with update frequency for typical off-road convoy operations.

13.2 Apply Radio Configuration

- Open the GroupTrack menu.
- Expand EVENT / RIDE.
- Tap Apply Radio (or Apply Convoy Config).
- The configuration is sent to your connected radio.
- Wait for confirmation that all 40 fields validated successfully.

Radio programming may require two passes. If any errors are returned on the first pass, run Apply Convoy Config a second time. All 40 of 40 fields must validate successfully. Do not proceed to a ride until the config shows 40/40 applied with no errors.

13.3 Archive and Restore

- Archive: backs up your current radio settings before applying convoy configuration. This preserves your personal radio settings so you can restore them after the ride.
- Restore: returns your radio to the archived settings after the ride is complete.

Always archive before applying convoy config, especially if your radio is configured for other uses outside of GroupTrack.

All riders in a convoy MUST be on the same frequency, channel, and modem preset. Use the Apply Radio feature to ensure consistent configuration across all riders. Mismatched settings prevent position sharing.

Chapter 14

Known Limitations (V2.4)

14.1 Current Limitations

- No mileage display: Trip distance / odometer is not calculated or displayed. Planned for V2.5.
- 60-second speed warmup: Speed shows 0 mph for the first approximately 60 seconds while the GPS acquires enough position samples for accurate speed calculation.
- GPS recording interval 3 seconds: Track points are recorded every 3 seconds. Higher resolution (1 second) is planned for V2.5.
- Budget device BLE: Low-cost tablets (Tab8NEU and similar) cannot hold BLE connection to T1000-E radios. The connection drops after 8-10 seconds. This is a confirmed hardware limitation.
- BLE reconnect: If BLE drops during a ride, manually disconnect and reconnect in the app. Keep your phone close to the radio (within 50 feet) to prevent drops.
- Battery optimization: Some Android devices kill background apps aggressively. Go to Settings > Battery > GroupTrack > Unrestricted to prevent this.
- Display panel position: The floating display panel position resets when navigating between screens. Position persistence is planned for V2.5.
- Track import naming: GroupTrack-recorded tracks include a _YYYYMMDD_HHMMSS suffix. Location-based naming is planned for a future release.
- Esri online map labels: Online maps may show sparse labels compared to downloaded offline tiles. This is due to Esri's deprecation of legacy tile services. Your offline maps retain the original dense labels.
- Remove Area: Tile removal via draw-to-delete region is coming in V2.5. Currently, tiles can only be removed by manually deleting directories via a file manager.
- Downloads in Progress overlay: Pink/red shadow visualization for active downloads and removals is coming in V2.5.

Chapter 15

Troubleshooting

Track file is empty (0 track points)

Location permission is set to "While using the app" instead of "Allow all the time." Fix: Go to Settings > Apps > GroupTrack > Permissions > Location > Allow all the time. This must be set BEFORE starting a recording.

Map is blank / no tiles visible

Offline tiles have not been downloaded for this area. Fix: Open the Planning Map, search for your area, use DOWNLOAD to cache tiles. Also confirm that All Files Access permission is granted in Settings.

Radio won't connect

Ensure the T1000-E radio is powered on (LED flashing) and within 50 feet of your phone. Try toggling Bluetooth off and on in your phone's Settings. If using a budget tablet, try a different Android device (see Appendix A for tested devices).

Radio connects then immediately disconnects

Usually caused by firmware version mismatch. Verify your T1000-E is on firmware 2.6.11. If the radio has been updated past 2.6.11, contact the GroupTrack team for downgrade instructions.

Radio configuration errors

Run Apply Convoy Config a second time. All 40/40 fields must validate. Two passes may be needed due to timing of radio parameter application.

Speed stuck at 0 mph

Wait 60 seconds after starting a ride. The speed display requires a warmup window to compute accurate speed from GPS samples. If speed remains at 0 after 2 minutes, check that location permission is set to "Allow all the time."

App killed in background

Go to Settings > Battery > find GroupTrack > set to "Unrestricted." This prevents Android from killing the app when the screen is off. Some manufacturers (Xiaomi, Huawei, Samsung) have aggressive battery optimization that must be disabled for background GPS recording.

Other riders not visible on map

Confirm all radios are running the same convoy config (same frequency, channel, modem preset). All riders must show as connected in the app. Position updates take a few seconds to propagate through the mesh network.

Download is slow or stalling

Use a strong WiFi connection for tile downloads. Cellular data works but is slower. Reduce the download zoom level for faster downloads (fewer tiles to fetch). The download queue processes 2 downloads simultaneously -- check the queue panel for status.

Downloads not visible on map

Toggle "Downloaded Areas" ON in the Display Panel to see blue shading where tiles are cached. If no blue shading appears, tiles may not have been downloaded for the current map view area.

Tracks not loading

The Tracks toggle in the Display Panel reads all files from Documents/my_tracks/. If the directory is empty, no tracks will appear. Import tracks using the TRACKS button on the Planning Map or record a ride to create new tracks.

Labels missing on Planning Map

On initial Planning Map load, labels may not appear immediately. Workaround: switch tile source (tap TOPO then SAT) to trigger a full tile refresh. Labels will appear after the source switch.

Chapter A

Hardware Compatibility

A.1 Tested Android Devices

GroupTrack has been tested and verified on the following devices. Other Android devices may work but are not officially supported.

- Android 1 (Samsung test device): stable BLE connection to T1000-E, reliable background GPS recording, all tile sources display correctly.
- Android 2 (Pixel test device): stable BLE connection, V3.0 development and testing device.

Budget Android devices (Tab8NEU tested on Android 10, 14, and 16) CANNOT hold BLE connection to T1000-E radios. Supervision timeout (reason 0x0008) occurs after 8-10 seconds. This is a confirmed hardware limitation of the Bluetooth chipset in budget devices.

A.2 Recommended Device Specifications

- Android 10 or later
- Bluetooth 4.2 or later (BLE support required)
- 2GB RAM minimum, 4GB recommended
- 16GB storage minimum for app + map tiles
- GPS with A-GPS support for fast position acquisition
- Name-brand manufacturer (Samsung, Google Pixel, OnePlus, Motorola)

Chapter B

T1000-E Radio Guide

B.1 Firmware Requirements

Your T1000-E MUST remain on firmware version 2.6.11. Do NOT accept firmware update prompts from the Meshtastic app or any other source.

Newer firmware versions (2.7.x and later) destabilize the BLE connection protocol and radio parameter configuration. Symptoms of wrong firmware include: immediate disconnection after pairing, failure to apply convoy configuration, and intermittent position update dropouts.

If your radio has been updated past 2.6.11, contact the GroupTrack team for downgrade instructions.

B.2 BLE Range and Placement

Keep your T1000-E in your cart at all times during rides. The radio must stay within 50 feet of your phone.

The T1000-E uses Bluetooth Low Energy (BLE) to communicate with your phone. BLE has a practical range of approximately 30-50 feet depending on obstructions. Exceeding this range causes the Bluetooth supervision timer to expire (reason code 0x0008), which disconnects the radio and disrupts position tracking.

Best practices for radio placement:

- Mount the radio in a visible location in your cart, not buried under gear.
- Keep your phone in the same cart as the radio.
- Do not carry the radio in a pocket or bag away from your phone during the ride.
- Metal cart frames can reduce BLE range -- mount the radio above the frame if possible.

B.3 Power Management

- Battery life: approximately 8-12 hours on a single charge depending on transmission frequency.
- Power on: press and hold the button for 3 seconds. LED will flash.
- Power off: press and hold the button for 5 seconds.
- Charge: USB-C charging. Charge fully the night before a ride.
- LED indicators: flashing = powered on and advertising, solid = connected, off = powered off or battery dead.

B.4 Mesh Radio Range

The T1000-E mesh radio range varies significantly by terrain:

- Line of sight: 0.5-2 miles depending on antenna orientation and terrain clearance.
- Canyon/forest: 0.25-0.75 miles with significant signal attenuation.

- Mesh relay: if riders are spaced within range of each other, messages relay through intermediate nodes, extending effective convoy range beyond direct radio range.

The LONG_FAST modem preset (recommended convoy configuration) optimizes for frequent position updates at moderate range. This is the best balance for typical off-road convoy operations.

Chapter C

Coming in V2.5

C.1 Planned Features

V2.5 is a 3-4 week development cycle introducing the spatial data engine -- a major expansion of GroupTrack's data management capabilities.

Waypoints

- Create waypoints via long-press on the map
- 10 waypoint types: trailhead, fuel, gate, hazard, scenic viewpoint, water source, camp, parking, rally point, other
- Rename, move, delete, and share waypoints with convoy riders
- Automatic map legend generated from active waypoint types
- Display panel Waypoints toggle becomes active

Routes

- Draw routes directly on the map
- Snap-to-trail routing following established paths
- Import routes from GPX and KML files
- Convert recorded tracks to reusable routes
- Assign routes to rides for convoy navigation
- Display panel Routes toggle becomes active

Trail Database

- Download full state trail datasets from public land management agencies
- On-device SQLite/GeoPackage storage with spatial indexing
- Filter trails by type: OHV, hiking, horse, multi-use
- Lazy loading based on map viewport -- only load trails visible on screen
- Smart tile downloads guided by trail coverage areas

Area Management

- Remove downloaded tiles for selected areas to free storage
- Draw-to-delete: draw a rectangle to remove tiles within that area
- Queue-based tile removal with progress tracking
- Red shadow visualization on map for areas being removed

Data Protection

- Automatic database backups on app launch and before destructive operations
- Change journal recording every data modification for audit trail
- Point-in-time rollback capability
- Manual backup and restore from Settings

Additional Map Sources

- Expanded tile source selection beyond SAT/TOPO/TOPO+
- Data-driven source catalog with up to 17 tile providers
- Per-source API key management

iPad Support

- Web-based companion app for iPad via Web Bluetooth browser
- Same map rendering and track display as Android
- Progressive Web App (PWA) deployment -- no App Store required initially

Chapter D

Glossary

AutoPan	Automatic map centering that follows your cart during recording. Off by default, activates during recording.
AutoPause	Automatic battery-saving pause when convoy is stationary for 10+ minutes.
BLE	Bluetooth Low Energy. The wireless protocol used to connect your phone to the T1000-E radio.
Convoy	A group of riders traveling together with position sharing via mesh radio.
GPX	GPS Exchange Format. Standard file format for GPS track data. Used by GroupTrack for track recording and import.
GeoPackage	SQLite-based spatial database format. Used in V2.5 for trails, waypoints, and routes.
KML	Keyhole Markup Language. Google Earth file format supported for track import.
Lead Cart	The rider at the front of the convoy, selected when starting a group recording.
LOCAL	Offline tile mode. Tiles loaded from downloaded cache on device.
Mesh Network	A radio network where each node relays messages, extending range beyond direct node-to-node distance.
Meshtastic	Open-source mesh radio platform used by GroupTrack for position sharing.
NET	Online tile mode. Tiles loaded from Esri servers via internet.
Planning Map	The map screen for downloading tiles, managing tracks, and planning rides. Formerly called Map Viewer.
PWA	Progressive Web App. A web application that can be installed on a device like a native app.
SAT	Satellite imagery tile source (Esri World Imagery).
T1000-E	A compact Meshtastic mesh radio used for convoy position sharing.
Tail Cart	The rearmost rider in the convoy, identified automatically by GroupTrack.
Tile	A small square map image (typically 256x256 pixels) at a specific zoom level and location.
TOPO	Topographic map tile source (Esri World Topo Map).
TOPO+	USGS topographic map tile source (Esri USA Topo Maps).
Track	A recorded GPS path saved as a GPX file. Represents a ride or route taken.
Trail	An established path or route from a public land management agency database.
Waypoint	A point of interest on the map (V2.5). Types include trailhead, fuel, gate, hazard, etc.
WebView	An embedded web browser component used by GroupTrack to render Leaflet maps.
WorkManager	Android background task system used for tile downloads that survive app backgrounding.